Continuous RVs: Expectation and Common Distributions

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MATH 241

Week 9, Lecture #2 — Ch. 5.2-5.5

Last Time and Today

- Continuous RVs
  - PDF, CDF
  - $E(X)$
- Today: common continuous RVs, applications

Reminders

1. Project proposal due tomorrow, 10/30 at 11:59pm via Gradescope
2. HW6 due Monday, 11/2 at 11:59pm
3. CP5 on Tuesday, 11/3
   - 20 minute quiz, open-note

Expectation (Definition 5.1.10)

The expected value of a continuous random variable X with pdf $f_X(x)$ is:

$$E(X) = \int_{-\infty}^{\infty} x\, f_X(x)\, dx$$

Question: What is $E(X^2)$? $E[g(X)]$?

LOTUS (Theorem 5.1.11)

If X is a continuous RV with a pdf $f_X(x)$ and g is a function from $\mathbb{R}$ to $\mathbb{R}$, then:

$$E[g(X)] = \int_{-\infty}^{\infty} g(x)\, f_X(x)\, dx$$

Note: $Var(X) = E(X^2) - E^2(X)$ can still be used for continuous RVs!

Back to Example #1

Suppose we care about the time until the next text message comes in from one of your family members, which is denoted as X. Say X has the following PDF:

$$f_X(x) = \frac{3}{2}e^{-\frac{3}{2}x} \qquad \text{for } x \geq 0$$

Objective: Find $E(X)$.

Example #1: Expectation

$$
\begin{aligned}
E(X) &= \int_{-\infty}^{\infty} x \, \frac{3}{2} e^{-\frac{3}{2}x} \, dx \\
&= \frac{3}{2} \int_{0}^{\infty} x \, e^{-\frac{3}{2}x} \, dx
\end{aligned}
$$

Example #1: Expectation

Let $u = x$, $dv = e^{-\frac{3}{2}x}dx$. Then $du = dx$ and $v = -\frac{2}{3}e^{-\frac{3}{2}x}$. So,

$$
\begin{aligned}
E(X) &= \frac{3}{2}\left[x\left[-\frac{2}{3}e^{-\frac{3}{2}x}\right]\Big|_0^\infty - \int_0^\infty \left[-\frac{2}{3}e^{-\frac{3}{2}u}\right] du\right] \\
&= \frac{3}{2}\left[x\left[-\frac{2}{3}e^{-\frac{3}{2}x}\right]\Big|_0^\infty - \frac{4}{9}\left[e^{-\frac{3}{2}x}\right]\Big|_0^\infty\right] \\
&= \frac{3}{2}\left[x\left[-\frac{2}{3}e^{-\frac{3}{2}x}\right]\Big|_0^\infty \to 0 - \frac{4}{9}\left[e^{-\frac{3}{2}x}\right]\Big|_0^\infty \to -1\right] \\
&= \frac{3}{2} \times \frac{4}{9} \\
&= \frac{2}{3}
\end{aligned}
$$

Activity

Let X be a random variable with the following PDF:

$$f_X(x) = e^{-x} \qquad \text{for } x \geq 0$$

1. Find the CDF of X.
2. Find $E(X)$.
3. Find $P(X > 2)$ and $P(X < 0.5)$.

Common Distributions

We will be looking at common distributions for continuous RVs. This includes the:

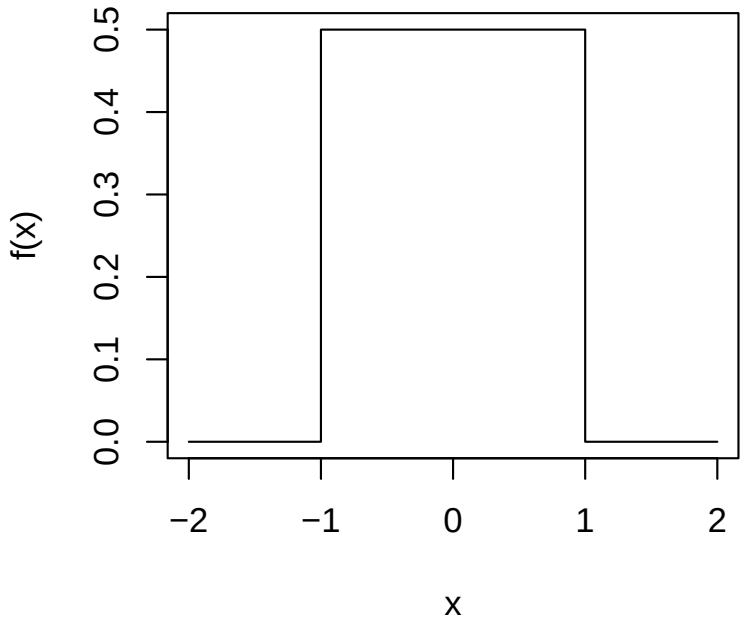
- Uniform distribution (i.e $X \sim Unif(a, b)$)
- Normal distribution (i.e $X \sim N(\mu, \sigma^2)$)
- Exponential distribution (i.e $X \sim Exp(\lambda)$)

Uniform RV - PDF

$$X \sim Unif(a, b)$$

- a and b are the endpoints
  - $a = 0$, $b = 1$: a standard uniform
- Constant (uniform) density between the endpoints

$$f_X(x) = \begin{cases} \frac{1}{b-a} & \text{for } a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$



Uniform RV - CDF

$$X \sim Unif(a, b)$$

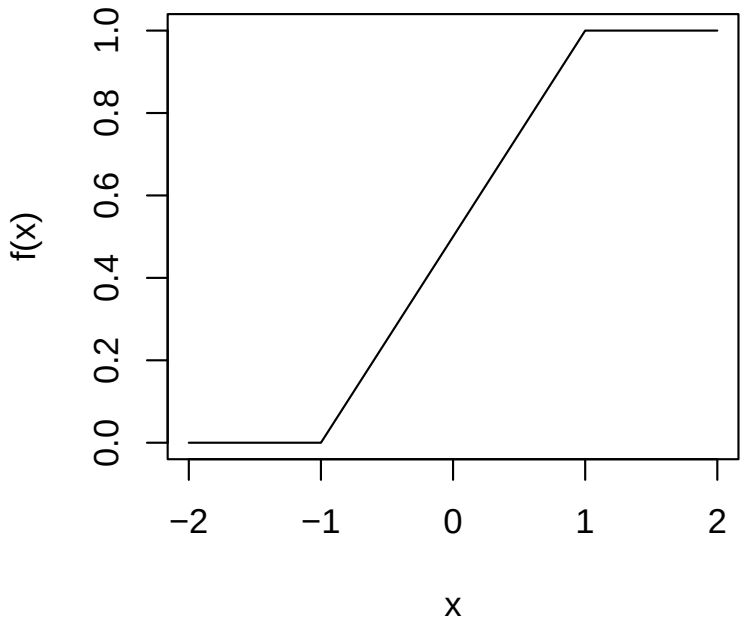
- What is the CDF?

Uniform RV - CDF

$$X \sim Unif(a, b)$$

- What is the CDF?

$$F_X(x) = \begin{cases} 0 & \text{for } x < a \\ \frac{x-a}{b-a} & \text{for } a \leq x \leq b \\ 1 & \text{for } x > b \end{cases}$$



Uniform RV - Expectation and Variance

Expectation of X:

$$E(X) = \frac{a+b}{2}$$

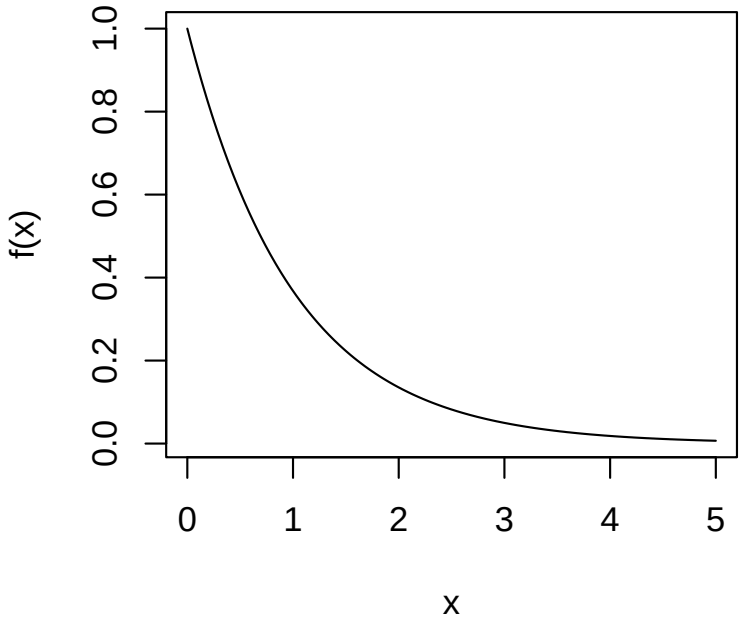
Variance of X:

$$Var(X) = \frac{(b-a)^2}{12}$$

Exponential RV - PDF

$$X \sim Exp(\lambda)$$

- λ is the rate of an event occurring (e.g. decay, flow)
  - $\lambda = 1$: the default
- Has memoryless property:
  $P(X > s + t \,|\, X > s) = P(X > t)$

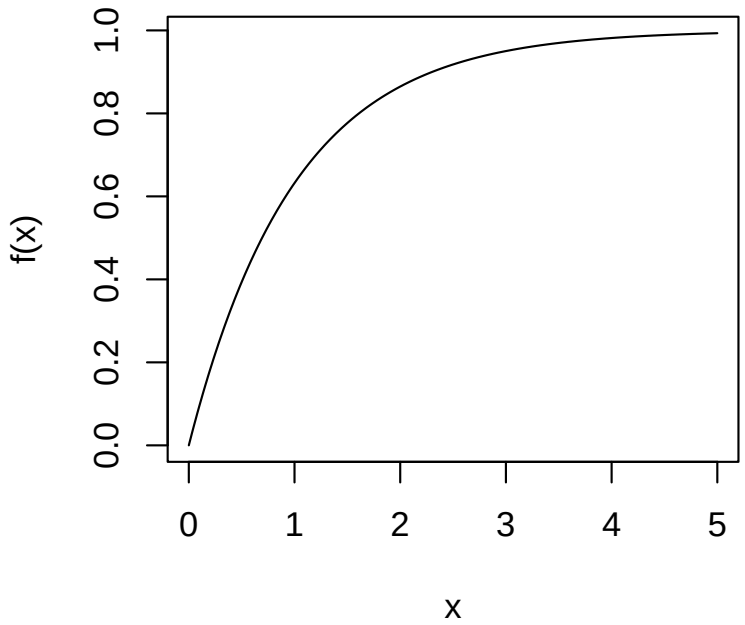


$$f_X(x) = \begin{cases} \lambda e^{-\lambda x} & \text{for } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

Exponential RV - CDF

$$X \sim Exp(\lambda)$$

- λ is the rate of decay/flow
  - $\lambda = 1$: the default
- Has memoryless property: $P(X > s + t \,|\, X > s) = P(X > t)$



$$F_X(x) = \begin{cases} 0 & \text{for } x < 0 \\ 1 - e^{-\lambda x} & \text{for } x \geq 0 \end{cases}$$

Exponential RV - Expectation and Variance

Expectation of X:

$$E(X) = \frac{1}{\lambda}$$

Variance of X:

$$Var(X) = \frac{1}{\lambda^2}$$

Note: $X \sim Exp(\lambda)$ may be reparameterized using $\beta = \frac{1}{\lambda}$, which is seen in some textbooks/resources.

Exponential RV - Example

Suppose the waiting time in line at the post office follows an exponential distribution, with expected waiting time of 5 minutes.

What's the probability that you will have to wait more than 5 minutes?

Exponential RV - Example

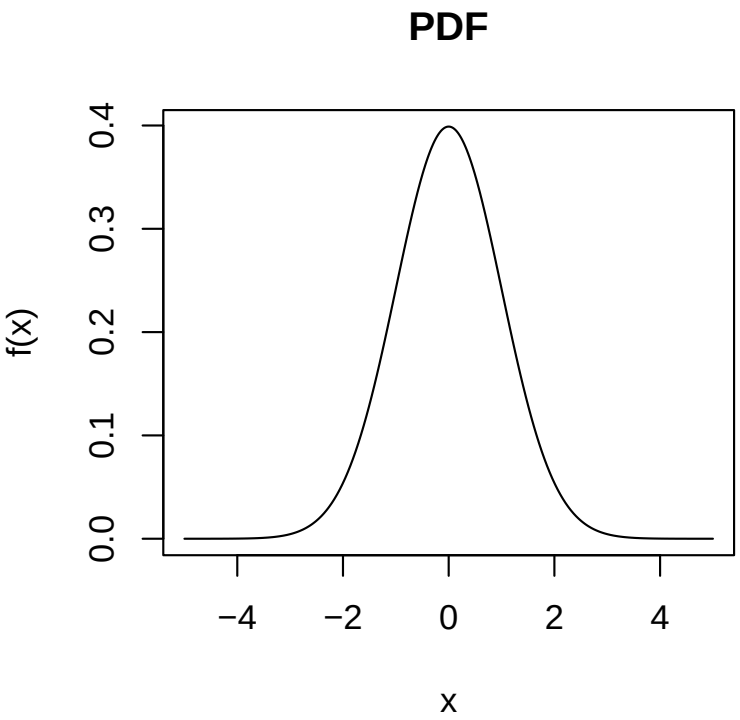
(1) Use $E(X)$ first to find λ:

$$E(X) = 5 = \frac{1}{\lambda} \implies \lambda = \frac{1}{5}$$

(2) Use the CDF of an exponential RV to find $P(X > 5)$:

$$\begin{aligned} P(X > 5) &= 1 - P(X \leq 5) \\ &= 1 - F_X(5) \\ &= 1 - \left(1 - e^{-\frac{1}{5}(5)}\right) \\ &= 0.368 \end{aligned}$$

Normal Distribution - Motivation



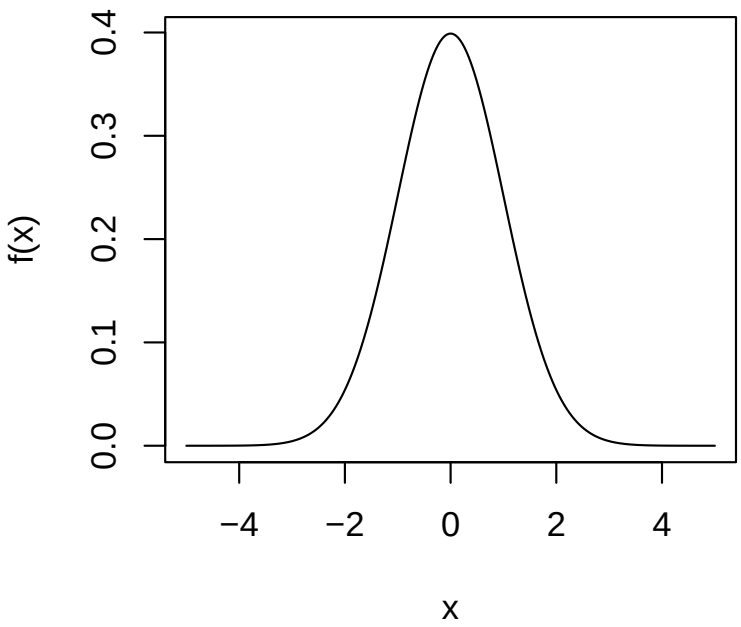
- Bell-shaped curve
- Used for **central limit theorem**, natural/physical processes (height, stock prices)
- Used for z-tests
- We start with the standard normal, followed by the normal

Standard Normal RV - PDF

$$X \sim N(\mu = 0, \sigma^2 = 1)$$

- Mean of 0 and variance of 1
- Used for z-scores
  - Empirical rule

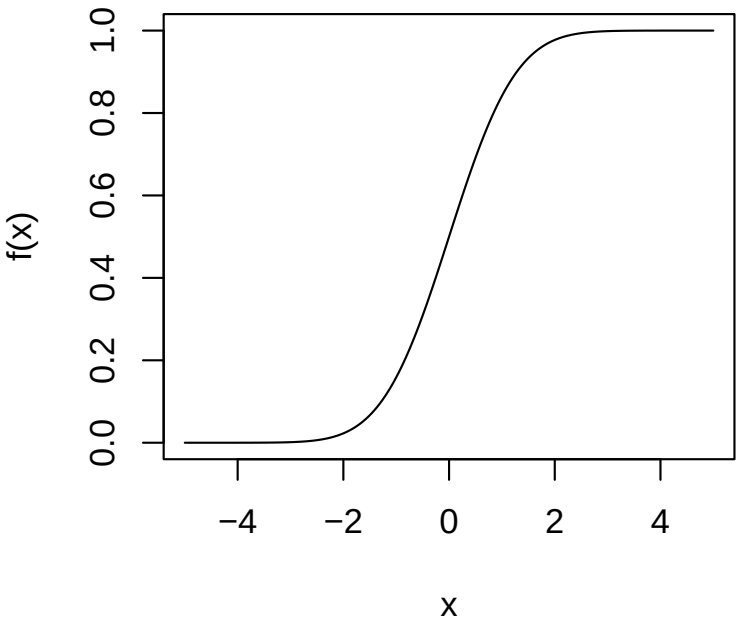
$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} \quad \text{for } -\infty < x < \infty$$



Standard Normal RV - CDF

$$X \sim N(\mu = 0, \sigma^2 = 1)$$

- CDF has no closed-form solution
  - Use of numerical approximations



Standard Normal RV - Expectation

Expectation of X:

$$E(X) = 0$$

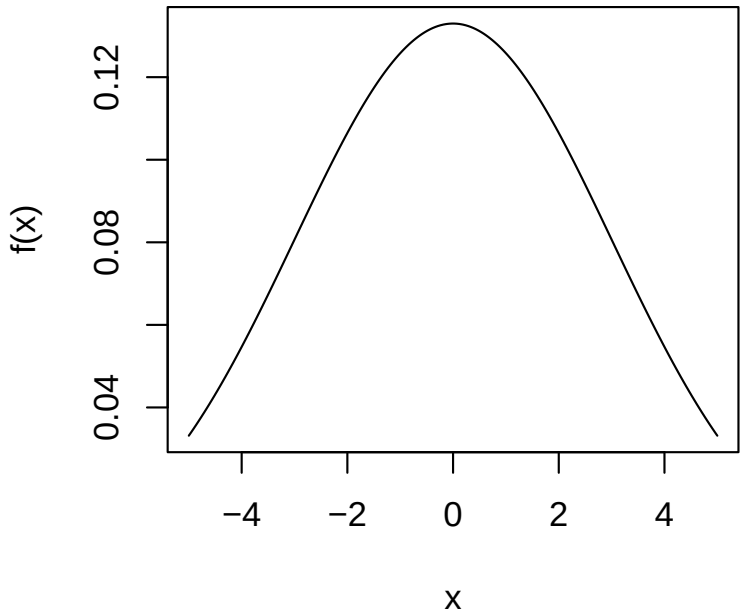
Variance of X:

$$Var(X) = 1$$

Normal RV - PDF

$$X \sim N(\mu, \sigma^2)$$

- Can revert to $Z \sim N(0, 1)$
  - $Z = \frac{X-\mu}{\sigma}$

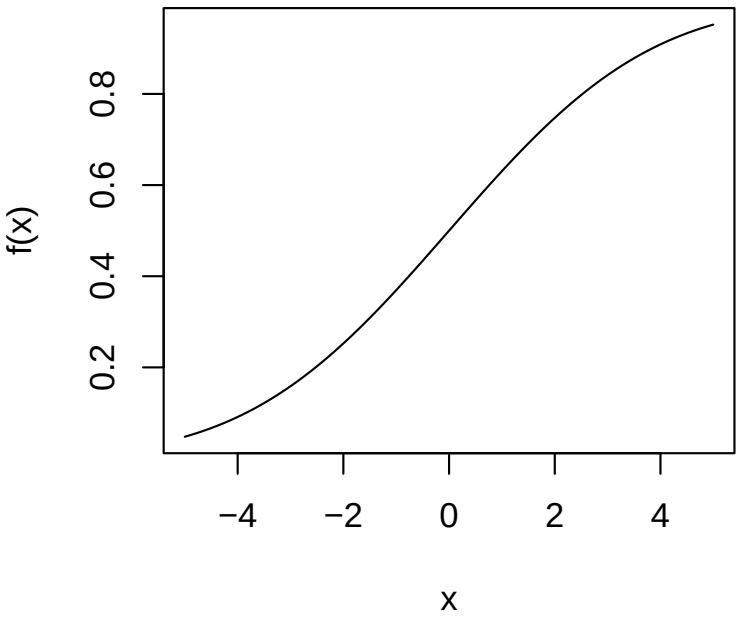


$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad \text{for } -\infty < x < \infty$$

Normal RV - CDF

$$X \sim N(\mu, \sigma^2)$$

- CDF has no closed-form solution
  - Use of numerical approximations



Normal RV - Expectation

Expectation of X:

$$E(X) = \mu$$

Variance of X:

$$Var(X) = \sigma^2$$

Application #1: z-tests and Empirical Rule

To go from $X \sim N(\mu, \sigma^2)$ to $Z \sim N(0, 1)$, we need to apply the following transformation:

$$Z = \frac{X - \mu}{\sigma}$$

- Provides us with how many SDs we are away from the mean
- Empirical rule: 68%, 95%, 99.7%

Application #2: Universality of the Uniform

Suppose X is a RV with strictly increasing continuous F on $\mathbb{R}$. For $0 < p < 1$, the unique value x_p such that $F(x_p) = p$ is called the pth quantile of the distribution. That is,

$$x_p = F^{-1}(p)$$

- To find $F^{-1}(p)$: isolate x from the equation for $F_X(x)$

Application #2: Universality of the Uniform

Let F be a CDF which is a continuous function and strictly increasing on the support of the distribution. This ensures that the inverse function F^{-1} exists, as a function from $(0,1)$ to $\mathbb{R}$. We then have the following results:

1. Let $U \sim Unif(0,1)$ and $X = F^{-1}(U)$. Then X is a RV.
2. Let X be a RV with cdf F. Then $F(X) \sim Unif(0,1)$.

- Can be used for inverse-transform sampling
  - Simulate $U \sim Unif(0,1)$
  - Find $F^{-1}(u)$
  - Substitute observations from U into $F^{-1}(u)$ to get random observations for X

Application #3: Normal Approximation to Binomial

If $X_n \sim Bin(n, p)$, then:

$$\frac{X_n - np}{\sqrt{np(1-p)}} \sim N(0, 1)$$

for large n. Equivalently, $X_n \sim N(np, np(1-p))$.